

PROFESSIONAL SUMMARY

Full-stack software engineer with production experience building enterprise applications using C#, ASP.NET Core MVC, JavaScript and MySQL. Work spans AI/LLM integration, database optimization, interactive dashboards and external system integration. Built and deployed a React, Node.js and MongoDB application for LLM emissions benchmarking.

TECHNICAL SKILLS

Languages	C#, JavaScript, SQL, HTML/CSS, Python	Databases	MySQL, SQL Server, MongoDB
Web	ASP.NET Core MVC, .NET 9, Razor, EF Core, Identity, React, Node.js, Express	AI and LLMs	Ollama, OpenAI-compatible APIs, structured actions, SQL generation
Visualization	DevExpress Dashboard, QGIS, choropleth maps, Power BI, Access	Integration	RabbitMQ, REST APIs, IIS, Git/GitHub, NuGet, SSL/TLS, Jira

WORK EXPERIENCE

ST Engineering | Software Engineer

AUG 2025 - PRESENT

Full-Stack Web Developer | Agile Team | Enterprise Applications

- Developed and maintained an enterprise web application using C#, ASP.NET Core MVC (.NET 9), Razor, JavaScript, DevExpress and MySQL; integrated SQL Server systems and RabbitMQ messaging.
- Designed an extensible AI assistant supporting Ollama and OpenAI-compatible APIs, configurable models, service abstractions and structured parsing of actions, SQL and responses.
- Optimized MySQL queries across datasets exceeding 800K records using EXPLAIN, indexing, views and query redesign.
- Built DevExpress dashboards for water and electricity monitoring with date filters, period comparisons, Top-N analysis, hourly averages and QGIS choropleth maps.
- Implemented ASP.NET Identity and role-based access control; connected multiple data sources using Entity Framework Core.

Jurong Port Pte Ltd | Project Support Intern

JUL 2023 - DEC 2023

Digitalization Roadmap | UAT | Reporting Tools

- Led UAT for digitalization projects by designing test cases, executing tests and documenting defects to support release sign-off.
- Built Power BI and Microsoft Access reporting dashboards to make operational data accessible to management.
- Coordinated vendors and internal teams, maintained requirements and tracked delivery through Jira.

PROJECT

LLM Carbon Footprint Tracker | NTU Final Year Project

AUG 2024 - MAY 2025

React | Node.js | Express | MongoDB | GitHub Pages | Render

- Designed and deployed a full-stack application for global LLM emissions benchmarking, owning requirements, implementation and deployment.
- Built a React interface with dynamic filters, sortable tables, interactive charts and a validated data submission form.
- Developed a Node.js/Express REST API with MongoDB and implemented an estimator for projected CO2 output from user-defined training parameters.

EDUCATION AND LEADERSHIP

Nanyang Technological University | BComp Computer Science, Honours

AUG 2021 - MAY 2025

(Distinction)

GEM Explorer Exchange Programme, The University of Hong Kong | Jan - May 2024

Basketball Team Captain | NTU Hall 6

2024/25 SEASON

- Captained approximately 15 players to the 2024/25 NTU Inter-Hall Basketball Championship title and represented the hall for three consecutive years.
- HackX! Hack for Public Safety 2024; Shopee Code League 2022.